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261100 690003 |

1. Decrypt the following cipher text using Hill cipher

i) CT: HLDTQI HJDX WQ CMAY, key $= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

Invers

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3\times7)(5\times2) = 21-10=11,\ 11^{-1} \bmod 26 = 19$$

$$P = K^{-1} c \bmod 26$$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26 = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3\times7+111\times11 \\ 4\times7+5\times11 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 175 \\ 118 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 7 \\ 14 \end{bmatrix} \bmod 26 = \begin{bmatrix} \overset{275}{3\cancel{\times7+111}} \\ 122 \end{bmatrix} \bmod 26 = \begin{bmatrix} 15 \\ 18 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \bmod 26 = \begin{bmatrix} 160 \\ 180 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \bmod 26 = \begin{bmatrix} 147 \\ 198 \end{bmatrix} \bmod 26 = \begin{bmatrix} 17 \\ 12 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix}\begin{bmatrix} 3 \\ 23 \end{bmatrix} \bmod 26 = \begin{bmatrix} 331 \\ 1112 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 12 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix}\begin{bmatrix} 22 \\ 16 \end{bmatrix} \bmod 26 \begin{bmatrix} 298 \\ 278 \end{bmatrix} \bmod 26 = \begin{bmatrix} 9 \\ 18 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}\begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 174 \\ 70 \end{bmatrix} \bmod 26 = \begin{bmatrix} 18 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}\begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 89 \\ 30 \end{bmatrix} \bmod 26 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

plaint text : TOP SECRET MESSAGE

2) Consider the Hill cipher key matrix $k = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$

cipher text = RXADDY. Find Plain text.

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = 241 \bmod 26$$

$$= \text{Rem } 9 \rightarrow 26 \times 9 = 234 - 241 = 7 \bmod 26$$

$$7^{-1} \bmod 26 = 15$$

$$15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

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$$\begin{bmatrix}2 & 1\\ 3 & 22\end{bmatrix}\begin{bmatrix}17\\ 23\end{bmatrix} = \begin{bmatrix}2\times17 + 1\times23\\ 3\times17 + 22\times23\end{bmatrix}$$

$$= \begin{bmatrix}57\\ 557\end{bmatrix} \text{md } 26 = \begin{bmatrix}5\\ 11\end{bmatrix} = \begin{bmatrix}F\\ L\end{bmatrix}$$

$$\begin{bmatrix}2 & 1\\ 3 & 22\end{bmatrix}\begin{bmatrix}0\\ 14\end{bmatrix} = \begin{bmatrix}14\\ 308\end{bmatrix} \text{md } 26 = \begin{bmatrix}14\\ 22\end{bmatrix}$$

$$= \begin{bmatrix}O\\ W\end{bmatrix}$$

$$\begin{bmatrix}2 & 1\\ 3 & 22\end{bmatrix}\begin{bmatrix}3\\ 24\end{bmatrix} = \begin{bmatrix}30\\ 537\end{bmatrix} \text{md } 26 = \begin{bmatrix}4\\ 17\end{bmatrix}$$

$$= \begin{bmatrix}E\\ R\end{bmatrix}$$

Plain txt = FLOWER

3. Confusion vs Diffusion

Confusion: It hided the the relationship between the plintxt & ciphr text by changing the value of charctrs

Diffusion :- It spreds the information of the plaintext by changing: It spreds the information of the plaintext by changing the position of charctrs

Cipher	changes char value	change character position	Confusion/ Diffusion.
Shift cipher	Yes	No	Confusion
Multiplicative cipher	yes	No	Confusion
Affine Cipher	yes	No	Confusion
Transposition Cipher	No	Yes	Diffusion

→ Relation ship between substitution & confusion

- Substitution change one character into another character. Therefore. substitution mainly provides confusion
- Transposition Encryption

Encrypt the message "The arrow is pointing north direction.

i) change key = 4

Th EA
RROW
IS PD
INTI
NGNO
RTHD
IREC
TION

Cipher text

TRIINRITHRSNGTRT
EOPINHEDAWDIODCN

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ii) Key = ATLAS

ATLAS
15324

ATLAS
15324
THEAR
RUWIS
POINT
NORTH
DIREC
TIONX

Cipher txt :-

TRPNDIAINTENEWIRRD
RSIHCXHDUUII

[Consider omp "pnd" nt "Aulyz"]

iii) Double
Key 1 = Break
Key2 = Lwn

BREAK
25413
THEAR
ROWIS
POINT
INGND
RTHDI
RECTT
OXX Xy

Cipher text :-
AINN DIIxTRPIR
BDRSTOIIIxEWIGH
CX HOUNTTEN

L W R
2 3 1
T H E
A R R
O W I
3 P D
I N T
I N G
N O R
T H D
I R I
T T E
O N X

Cipher :- ERTUTGRDEI XTAU3I
IINT ILOHRLWPNNUHRTN

5. Transposition Decryption

i) Ciphertext:- TI DTE MRNME REUHO SA NUW
RSLTP EEFOD SSNU AHJ UT BIEND
GORAE PEJXD ELNSY = 60/10 = 6

key = 10

T R O O P S T N E E
I N G W E S V P P L
D M U R E N D H E N
I E S S E U B O I S
E R A Z D A I D X Y
M E N T U H E A O Y

Plain text: = Troops HEADING WEST
NEED MORE SUPPLIES AND
GENERA DUBOIS MEN TO AID